

KARS, Turkey

WMO: 170980

Lat: 40.564N

Lon: 43.112E

Elev: 1795

StdP: 81.54

Time Zone: 3.00 (E03)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.8	-22.1	-27.0	0.4	-24.6	-24.2	0.5	-21.9	12.1	-1.9	10.7	-2.3	0.6	300	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.2	29.9	15.8	28.0	15.5	26.2	15.3	18.2	26.0	17.1	24.8	16.3	24.0	5.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	13.4	22.4	14.1	12.6	21.3	13.2	11.8	20.0	59.3	26.0	55.8	24.8	52.8	23.9	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.3	8.0	-27.5	32.3	3.7	1.9	-30.2	33.6	-32.4	34.8	-34.5	35.8	-37.2	37.2		
(5)			-27.6	19.8	3.7	1.7	-30.3	21.1	-32.5	22.1	-34.6	23.1	-37.3	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-10.1	-7.9	-0.9	5.5	10.2	14.3	17.8	18.5	13.9	8.1	0.8	-6.7
(7)		DBStd	10.55	5.65	5.82	5.21	3.51	2.69	2.45	2.47	2.53	2.97	3.24	4.37	6.09
(8)		HDD10.0	2504	622	502	338	139	29	1	0	0	4	74	277	518
(9)		HDD18.3	4791	881	735	596	384	251	122	38	29	134	317	527	776
(10)		CDD10.0	815	0	0	0	4	37	130	243	262	123	16	0	0
(11)		CDD18.3	59	0	0	0	0	0	1	23	33	2	0	0	0
(12)		CDH23.3	1244	0	0	0	0	2	74	440	586	141	1	0	0
(13)		CDH26.7	317	0	0	0	0	0	6	121	169	22	0	0	0
(14)	Wind	WSAvg	2.8	1.9	1.9	3.3	3.5	3.2	3.1	3.4	3.5	2.9	2.5	2.2	1.8
(15)	Precipitation	PrecAvg	509	21	22	29	52	79	78	57	42	29	42	26	23
(16)		PrecMax	759	54	61	61	123	160	160	149	102	75	113	81	75
(17)		PrecMin	322	1	0	2	12	28	18	3	5	1	1	0	2
(18)		PrecStd	117	13	12	14	26	25	36	33	27	21	26	18	14
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.1	6.3	14.8	20.3	22.8	27.2	32.0	32.2	29.1	22.2	14.3	9.7
(20)		MCWB	0.8	3.3	6.6	9.1	12.2	14.8	16.7	16.2	13.7	10.8	6.2	3.6	
(21)		2%	DB	1.5	3.2	11.2	17.2	20.9	25.3	29.9	30.2	26.8	20.1	12.2	5.1
(22)		MCWB	-0.1	1.0	5.0	8.4	11.6	14.7	16.3	15.8	13.1	10.2	5.5	1.4	
(23)		5%	DB	0.1	1.9	8.9	15.2	19.0	23.8	27.8	28.7	24.9	18.0	10.1	3.0
(24)		MCWB	-1.2	0.1	4.1	7.7	10.7	14.4	16.1	15.7	13.1	9.7	4.7	0.5	
(25)		10%	DB	-1.1	0.2	6.2	13.0	17.2	22.0	25.9	26.9	23.0	15.9	8.0	1.2
(26)		MCWB	-2.3	-1.3	2.8	6.8	10.2	13.8	16.1	15.7	12.6	8.9	3.9	-0.6	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.3	3.4	7.7	10.7	14.1	17.6	20.0	20.0	16.6	12.8	7.6	3.9
(28)		MDCB	2.3	5.7	12.6	16.2	20.8	24.2	28.3	28.6	25.1	20.2	12.0	7.5	
(29)		2%	WB	0.1	1.3	5.7	9.3	12.6	16.4	18.4	18.2	15.1	11.3	6.4	1.9
(30)		MDCB	1.0	2.8	10.2	15.5	18.5	23.1	26.1	26.6	23.6	17.9	10.2	4.2	
(31)		5%	WB	-1.1	0.0	4.3	8.3	11.7	15.3	17.3	17.1	14.0	10.2	5.3	0.7
(32)		MDCB	0.0	1.6	7.7	13.9	17.3	22.0	24.7	25.4	22.4	16.4	9.0	2.6	
(33)		10%	WB	-2.5	-1.1	3.0	7.2	10.8	14.3	16.5	16.2	13.0	9.2	4.3	-0.4
(34)		MDCB	-1.3	0.1	6.2	12.3	16.1	20.6	23.8	24.5	21.0	14.7	7.6	1.1	
(35)	Mean Daily Temperature Range	MDBR	10.6	11.4	10.5	12.2	12.6	14.4	15.3	16.2	16.5	13.8	12.5	10.8	
(36)		5% DB	MCDBR	8.7	10.1	13.4	16.2	15.8	17.1	18.3	18.5	19.6	17.8	15.9	12.6
(37)		MCWBR	7.1	8.0	7.9	8.6	8.1	8.1	7.5	7.0	8.5	9.2	9.8	9.0	
(38)		5% WB	MCDBR	7.2	9.0	11.4	14.1	13.8	15.5	16.0	16.3	17.6	15.1	13.8	10.6
(39)		MCWBR	6.1	7.3	7.2	8.0	7.8	8.4	7.8	7.9	8.6	8.2	8.9	7.9	
(40)	Clear-Sky Solar Irradiance	taub	0.261	0.292	0.341	0.393	0.371	0.349	0.382	0.375	0.323	0.329	0.282	0.259	
(41)		taud	2.312	2.235	2.233	2.158	2.314	2.403	2.330	2.340	2.468	2.405	2.480	2.405	
(42)		Ebn at Noon	906	919	907	878	906	923	888	886	919	870	880	884	
(43)		Edh at Noon	87	111	125	145	127	117	125	120	98	93	74	74	
(44)	All-Sky Solar Radiation	RadAvg	2.53	3.70	4.67	4.95	5.65	6.88	7.03	6.41	5.20	3.40	2.30	1.90	
(45)		RadStd	0.16	0.25	0.46	0.48	0.39	0.44	0.37	0.25	0.32	0.26	0.29	0.23	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.94	-0.94	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		N/A	N/A	N/A	N/A	-0.47	N/A	N/A	N/A	+73	N/A

Nomenclature: See separate page